Sakthivel Vasudevan

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Professional Summary

Accomplished Senior R&D Engineer with nearly 10 years of hands-on expertise in product development, vehicle integration, and VA/VE cost-reduction initiatives for off-road vehicles, tractor platforms, and farm implements. Proven track record across top-tier OEMs (TAFE, JCB India, Mahindra & Mahindra) leading New Technology Innovation (NTI) and New Product Introduction (NPI) projects, including hybrid and electric vehicle (EV) tractor systems. Skilled in end-to-end design lifecycle management.

Skills & Abilities

- Design Calculation for structural
- Hydraulic Schematic Circuit
- Stack-up Analysis
- NPI/NTI Project Management
- DFMEA / DFM / DFA / DFS
- CREO / UG NX
- FEA - CREO simulation
- Windchill / Team center PLM

Experience

SENIOR MEMBER | TAFE LTD. | JUNE 2024 – PRESENT.

- Managing NTI and NPI projects for hybrid and electric vehicle (EV) tractor platforms.
- Designing next-generation tractor concepts for high-horsepower (> 75 HP) cabin and hybrid models.

ASSISTANT MANAGER | JCB INDIA LTD. | NOVEMBER 2023 – MAY 2024.

- Handled Structures & Hydraulic systems for UK Loadall (Telehandler) & RTFL Projects.

DESIGN ENGINEER | MAHINDRA & MAHINDRA | OCTOBER 2022 – PRESENT.

- Handled sub system design of chassis system like steering components.
- Handled NPI project & preparing SOR, DVR, Design documents, Field visit.

ENGINEER | TAFE LTD. | MAY 2017 – OCTOBER 2022.

- Handled Vehicle integration of various aggregates like Throttle, clutch & brake controls system, Front weight frames, Fuel system & sheet metal components for NPT / CI projects in **Tractor Division**.
- Concept generation & Implements packaging for Structural Products (Chassis frame) like Rev. Mould Board Plough Mini variant (45HP Range), Round Baler & Rake-11Arms Projects in **Implement Division**.

DESIGN ENGINEER | IND-AUSSIE SOLAR PVT LTD | MAY 2016 – MAY 2017.

- Creating a new concept for solar panel mounting structure.
- Preparing 3D models & 2D drawings for proto model - testing purpose.

Role in TAFE (Advance Engineering Dept.):

- Conducting global and domestic (India) benchmarking and system-level competitive studies.
- Preparing benchmarking reports and presenting strategic design insights to top management.
- Finalizing system proposals based on design review outcomes and engineering feasibility.
- Preparing Scope of Requirements (SOR) documentation to define project boundaries during initiation phases.
- Drafting Concept Brief Request Forms (CBRF) and leading vendor onboarding for Proof of Concept (POC) projects.
- Creating high-lift capacity concepts and electrically operated spool valve designs for advanced **EV tractor** platforms.
- Developing new engineering concepts for Digital Power Steering (DPS) integrated with GPS auto-guidance and headland management for **hybrid and >75HP cabin tractors**.
- Designing system architecture layouts for NTI projects, focusing on DPS with auto-guidance and automated headland management.
- Conducting product and project demonstrations with the CTO and senior management to align on future roadmaps and strategic next steps.
- Supporting product and feature readiness for major global trade fairs, including "**Agritechnica**" (Germany) and "**EIMA**" (Italy).

Role in JCB:

- Developed concept designs for structural assemblies and hydraulic systems in Telehandler and RTFL products.
- Verified 3D models and 2D drawings for engineering design releases.
- Designed the hydraulic circuit schematic for JCB 940 RTFL product platform.
- Prepared design review documents and presented concepts to the UK design lead for stage-gate approvals and drawing releases.

Role in M&M:

- Handled subsystem design for hydraulic and steering components.
- Prepared SOR and design documents for new and updated components.
- Performed design calculations, including pump flow, TCR, and kingpin torque.
- Coordinated with the CAE team for FEA analysis and worked with the CFT to close ORCs.
- Created 3D installation models and 2D drawings while reviewing DFA and DFS guidelines.
- Conducted field visits for new product performance trials.
- Coordinated with materials procurement teams to secure parts for prototype builds.

Role in TAFE (Tractor & Heavy Machinery implements):

- Handled Vehicle integration of various aggregates like Throttle, clutch & brake controls system, Front weight frames, Fuel system & sheet metal components NPT/ CI projects in tractor platform.
- Concept generation & Implements packaging for Structural Product (Chassis frame) like Rev. Mould Board Plough Mini variant (45HP Range), Round Baler & Rake-11Arms Projects in Implement Division.

- Benchmarking study for NPT Projects & Field initial trail support to prove the performance of implements.
- Preparing SOR for new & updating components & Preparing Design document & Field validation reports.
- Knowledge in packaging design for various aggregate.
- Knowledge in Sheetmetal sub components like drawbar, front weight frame & mounting brackets.
- Knowledge in brake & clutch system components like pedals & linkages.
- Participating on cross-functional project teams to perform design tasks, support builds & working for Cost reduction projects also.
- Good knowledge & Hands on experience Design calculation (like FOS for Sheetmetal components, linkage calculation as well as structural calculation also), clearance checks, DMU, Tolerance stacks, and other engineering analysis (FEA).
- Addressing & providing solution for field failure parts & Improve design based on FFR study.
- Good knowledge in manufacturing (Sheetmetal) process as well as IS & ISO standards also.
- DFM & DFA to ensure designs manufacture ability & its assembly.
- Preparation of DFMEA for identify early potential failure modes for a product.
- Tracking feedback from testing and service, report failure & providing corrective actions. Analyses manufacturing and field problems for root cause, then recommend and implement solutions.
- Good knowledge in CAD 3D & 2D software as well as CREO simulation also.
- Preparing 3D models & 2D drawings for ECR/ECN release process & Ensure EBOM release related activity.
- Handled in cost reduction projects also.
- Projects in **Tractor division**: TAFE 42 DI, MF240 SN.

In Implements Dept.: -

- Design of structural parts for **RMBP MINI** Variant under 45HP Tractors.
- Based on Field Failure Report, creating a new concept of "Single Point Cutting Tool" for Penetration of soil.
- Preparing Load calculation (FOS), FEA analysis in CREO simulation.
- Based on FEA analysis, improving Leg strength & Assy mainframe for both Mini & Regular Variant V 2.0 – **RMBP** Product.
- Supporting Field initial trail to prove the performance of implements & strength aspect also.
- **Round Baler** Machine: Based on benchmarking data to create new concept for Roller-Split Type & Frame Aggregate and preparing design documents for review.
- **RAKE-11 Arms** Machine: Based on benchmarking data to create new concept for Arms Aggregates (Entire product).
- Initially handled this project as Trade& Tieup category. Then I did reverse engineering the M/s Minos Rake (Entire product). Made 3D model & packaging all aggregates. Discussed with M/s Comer industry for Gear box bought out.
- In Product launch, I demonstrated Rake Product to Sales & Service team & Initial training program also conducted in Product Training Center.

Role in IND-AUSSIES Solar Prv. Ltd.:

- Creating a new concept for solar panel mounting structure on the train coach & fitting 4.8kW solar panels in top of the Train Coach with self-power of all fans & lights. This is the first coach in Southern zone.

- Design & Fabrication mounting structure of solar panels depends on the site condition.
- Design a mounting and Array structure for kW and MW projects as well as Industry & Home site.
- Preparing 3D models & 2D drawings for proto model - testing purpose.
- Preparing shortage list and material requirements for the project based on design.

S NO	PROJECTS NAME	CAPACITY
1.	Annamalai University, Chidambaram.	200kW
2.	Solar on coach (Fixed Top of the Coach roof)	4.8kW

Education

- B.E (AUTOMOBILE) | CGPA:7.7 | IRTT, ERODE | 2012-2016
- 12TH | 92.33% | BHARANI PARK HR. SEC. SCHOOL, KARUR | 2010-2012
- 10TH | 91.4% | MORNING STAR HIGH SCHOOL, KARUR | 2002-2010

Declaration:

I hereby declare that the above furnished details are true to the best of my knowledge.

Place : Chennai. Signature

Date : (SAKTHIVEL V)